2026

# Artefact 2 Report: Hybrid AI Prediction and Inference System

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## Capstone and Team Integration Statement

Our team is developing a hybrid AI system to predict periods of electricity-network stress in NSW using demand, renewable generation, weather, and grid data. My contribution for Artefact 2 involves building a machine learning and inference sub module which will predict electricity grid network stress one-hour in advance. The network stress can either be classified as low, medium, high or critical and based on this prediction, a corresponding recommendation will be provided. This prediction will be one of the key inputs used in the overall hybrid inference system.

## Introduction

### Problem Statement and Objectives

NSW's electricity system is expected to undergo significant transformation over the coming decades, driven by the growing demand from AI data centres and the transition towards renewable energy [1]. As a result, predicting periods of electricity network stress is becoming increasingly important. However, predictions alone are insufficient without appropriate mitigation strategies [2].

Therefore, the objective of our hybrid AI system is two-fold: analyse electricity demand, renewable energy generation, and grid conditions to predict periods of network stress and recommend appropriate actions to grid operators.

Artefact 1 focused on analysing, cleaning, and preparing the OpenElectricity dataset, which serves as the primary data source for the project. Building on this, Artefact 2 focuses on developing a hybrid AI system that combines machine learning models for predicting network stress with a symbolic AI layer that interprets these predictions and generates recommendations.

## Hybrid System Rationale

This hybrid AI approach is suitable for our project objectives as it combines the predictive capabilities of machine learning algorithms with the rule-based reasoning of symbolic AI. This allows the system to identify complex patterns in electricity data and generate recommendations based on explicit, predefined rules. This is particularly beneficial for grid operators, our primary stakeholders, as it provides transparency into how recommendations are generated, helping them make informed decisions while keeping a human-in-the-loop.

Our hybrid AI system consists of two main components:

- Machine Learning Component: Four machine learning models are used to predict electricity demand, network stress, renewable energy sufficiency, and electricity price spikes. Each model provides predictions based on patterns identified in historical electricity data.
- Symbolic AI Component: The symbolic layer takes the outputs of these four models and applies predefined facts and logical rules to assess overall network stress and generate corresponding mitigation recommendations.

# Dataset and Modelling

## Data Acquisition and Cleaning

The pipeline retrieves 234,312 hourly observations from the OpenElectricity dataset for NSW, covering 1 January 2000 to 24 September 2026. Features include electricity price, demand, renewable generation, curtailment, imports/exports, and network measurements.

Data cleaning validates numeric types, timestamps, duplicates, missing values, and hourly continuity. Unusual values, including negative curtailment and renewable proportions exceeding 100%, are flagged for investigation. A factor-of-1,000 correction is applied to the *generation_renewable_energy* field to address an identified scaling error in 2000.

## Feature Selection and Engineering

Twelve redundant or unsuitable columns are removed, including duplicated energy measurements, market value, and battery storage, leaving 14 original columns. Internal gaps of up to three hours are interpolated and identified using a *was_imputed* flag.

Cyclical calendar features are generated using sine and cosine transformations to capture daily, weekly, and seasonal patterns. Additionally, lagged values, 24-hour rolling means, and standard deviations are calculated for price, demand, renewable generation, and curtailment, producing 20 historical features.

## Scaling, Dimensionality Reduction, and Modelling

Nine input features are standardised before applying Principal Component Analysis (PCA) for dimensionality reduction and K-means clustering to identify similar observations. Isolation Forest detects 1,056 anomalies (approximately 2%) and classifies 51,744 observations as normal.

## Data Assessment Strategy

Data quality is assessed through structural validation, descriptive statistics, missing-value analysis, and outlier detection. The pipeline saves datasets at each preprocessing stage in Parquet format, alongside JSON reports documenting transformations and data-quality issues to support reproducibility.

# HYBRID SYSTEM ARCHITECTURE AND IMPLEMENTATION

## DESIGN

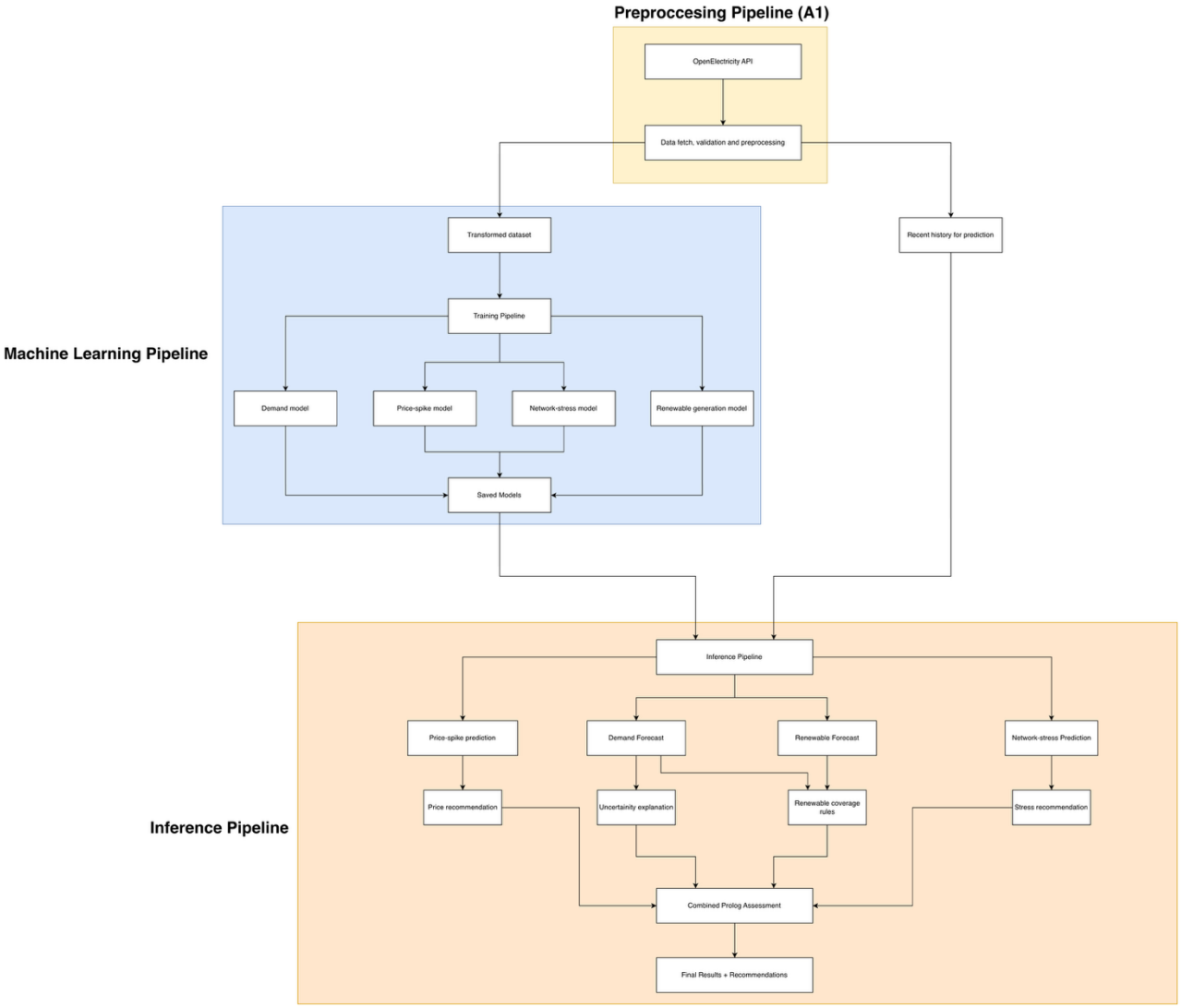



*Figure 1: Hybrid System Architecture*

## TRAINING PIPELINE

For the network-stress subsystem, the pipeline prepares the prediction target, compares two classification algorithms, evaluates the best-performing model, and saves the trained model for future inference.

The network-stress target is divided into four categories: low, medium, high, and critical. Due to the lack of actual engineering measurements in the dataset, network stress is estimated using proxy indicators related to network adequacy and energy congestion. Each measurement is assigned severity points based on the 75th, 90th, and 97th percentiles calculated from the training data. These points are combined into weighted scores, with the higher score determining the stress category. The target is then shifted backwards by one hour, allowing measurements at hour t to predict network stress at hour t + 1.

The dataset is divided chronologically into 60% training, 20% validation, and 20% testing. Rows are assigned based on their target timestamps, and percentile thresholds are calculated using only the training period to prevent data leakage.

The pipeline compares two tree-based classification algorithms:

- Random Forest: Provides a reliable baseline by combining predictions from multiple decision trees. Implemented using scikit-learn with 500 trees, a minimum of five samples per leaf, and balanced class weights within each bootstrap sample.
- XGBoost: Tests whether building decision trees sequentially, with each new tree correcting previous errors, can improve prediction accuracy. Implemented with 500 boosting rounds, a learning rate of 0.05, a maximum tree depth of six, and row and column sampling fractions of 0.8. Balanced sample weights are also used to address class imbalance.

Both models use class-balancing techniques. The model achieving the highest validation Macro-F1 score is selected, as this metric treats all four stress categories equally. Accuracy and critical-class performance are also evaluated. The selected algorithm is retrained using the combined training and validation datasets before being evaluated on the held-out test set.

## Symbolic AI System

The network-stress subsystem uses Prolog for symbolic reasoning and generating recommendations. Python runs the trained machine learning model to predict the next-hour network-stress level, while PySwip provides an interface between Python and the Prolog knowledge base.

For each prediction, three runtime facts are passed to Prolog:

- Predicted risk: The predicted network-stress category for the next hour (low, medium, high, or critical).
- Import share: The proportion of current gross electricity demand supplied by imports.
- Curtailment: The current amount of curtailed electricity generation, with negative values treated as zero.

Using these facts, the Prolog knowledge base identifies two possible causes of network stress:

- Adequacy: Identified when the import share is at least 0.15 (15%), indicating a significant reliance on imported electricity.
- Congestion: Identified when curtailment reaches at least 250 MW, suggesting that potential network constraints may require investigation.

The knowledge base then combines the predicted stress category with the identified stress drivers to generate recommendations based on predefined rules.

*Table 1: Prolog recommendation rules*

| **Condition** | **Recommendation** |
|---|---|
| Low predicted stress | Continue monitoring. |
| Medium predicted stress | Review forecasts. |
| High stress with an adequacy driver | Investigate supply conditions. |
| High stress with a congestion driver | Investigate network constraints. |
| Any high predicted stress | Review high-risk conditions. |

| | |
|---|---|
| Critical predicted stress | Notify the system operator. |

A final Prolog reasoning layer combines the network-stress prediction with outputs from the other subsystems, including demand uncertainty, electricity price predictions, renewable coverage, and data quality.

The system uses predefined rules to determine the overall response:

- Incomplete evidence: Requires manual review due to missing or unreliable information.
- Critical network stress: Requires urgent review by the system operator.
- Other warnings: Conditions such as high network stress or high demand uncertainty require further investigation or increased monitoring.

The final output provides an overall assessment of the predicted network conditions, including the triggered Prolog rules, explanations for the decisions, and recommended actions. In turn, this allows the hybrid AI system to combine machine learning predictions with explainable, rule-based reasoning to support electricity-network monitoring and decision-making.

## Evaluation and Analysis

### Model performance

The network-stress sub module evaluates and compares XGBoost and Random Forest (RF) using validation and test datasets, with results shown in Figure 2.

```
Validation performance:
               Accuracy  Macro-F1  Critical recall
Random Forest    0.7639    0.7010           0.7224
XGBoost          0.7833    0.7294           0.7302
Selected model: XGBoost

Test performance:
         Accuracy  Macro-F1  Critical recall
XGBoost    0.8127    0.7554           0.8952
Scores are on a 0–1 scale; higher is better.

Test performance by class:
precision    recall  f1-score   support

        low       0.94      0.88      0.91      5299
     medium       0.66      0.75      0.70      2389
       high       0.58      0.52      0.55      1222
   critical       0.83      0.90      0.86      1650

   accuracy                           0.81     10560
  macro avg       0.75      0.76      0.76     10560
weighted avg      0.82      0.81      0.81     10560
```

*Figure 2: XGBoost and RF model performance*

On the validation dataset, XGBoost outperformed RF, achieving an accuracy of 0.7833, a Macro-F1 score of 0.7294, and a critical recall of 0.7302. Macro-F1 was the primary metric for model selection as it treats all stress classes equally, regardless of frequency. This is important for reliably identifying different levels of network stress.

On the test dataset, XGBoost achieved an accuracy of 0.8127 and a Macro-F1 score of 0.7554, showing improved performance on unseen data. The model performed well when identifying low-stress observations, with 94% precision, and critical-stress observations, with 90% recall. However, it struggled with high-stress observations, correctly identifying only 52%.

Overall, XGBoost provides a useful predictive component for the hybrid system, although improvements could be made to better identify high-stress conditions.

## HYBRID SYSTEM DEMONSTRATION

Figures 3–5 demonstrate the system's operation, from model training and prediction to symbolic inference and final recommendations. The inference process was tested using the timestamp 3 November 2026 at 10:00 PM**.**

```
========== Network Stress | Training ==========
Candidates: Random Forest, XGBoost | Horizon: 1 hour
Selection: Highest validation macro-F1

Validation performance:
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Saved model: /Users/hanskonradpujalte/Documents/COMP3742-Capstone-Project/A2/saved_models/network_stress_model.joblib
Saved metadata: /Users/hanskonradpujalte/Documents/COMP3742-Capstone-Project/A2/saved_models/network_stress_model_metadata.json
Status: Training and evaluation complete

========== Renewable Generation | Training ==========
Model: Histogram Gradient Boosting | Horizon: 1 hour
Configuration: Previously selected in the renewable investigation notebook
Training examples: 52,631
Training targets through: 2026-09-23T23:00:00+10:00

Evaluation: Not performed in this training method.
See renewable_investigation.ipynb for the recorded model comparison.
```

*Figure 3: Machine learning sub modules training*

```
========== Demand Forecast ==========
Forecast for: 2026-11-03 22:00 +1000
Assessment: 7,433 MW
  95% interval: 7,311–7,555 MW
  Uncertainty: Low
Explanation:
  - this hour has historically been easy to predict (overnight, stable load)
Recommendations:
  - See the combined assessment below.

========== Network Stress ==========
Forecast for: 2026-11-03 22:00 +1000
Assessment: Low
Explanation:
  - Potential drivers from current observations: None identified
Recommendations:
  - Continue monitoring

========== Price Spike ==========
Forecast for: 2026-11-03 22:00 +1000
Assessment: Normal price
  Spike probability: 0.0% | Decision threshold: 28.0%
  Data quality: Valid | Confidence: High
Explanation:
  - [r4_normal_operations] Normal price expected and grid demand is normal; continue the standard schedule.
Recommendations:
  - Maintain operations

========== Renewable Coverage ==========
Forecast for: 2026-11-03 22:00 +1000
Assessment: Insufficient against the renewable-share target
  Generation: 1,725 MW
  Coverage: 23.2% | Target: 50.0%
  Target shortfall: 1,991 MW
  Demand sensitivity: Below across demand band
Explanation:
  - Forecast renewable generation 1725.1 MW covers 23.2% of forecast demand 7432.8 MW. The project target is 50% (3716.4 MW). Renewable target shortfall: 1991.3 MW.
  - Hourly renewable coverage of demand; total supply reliability is not assessed.
  - Demand-band sensitivity holds renewable generation fixed. Renewable forecast uncertainty is not quantified.
Recommendations:
  - Review available renewable supply and renewable-charged storage before considering a response.
  - Consider shifting flexible demand to hours with higher renewable availability.
  - Check other supply and reserves separately; a renewable target shortfall alone does not imply a blackout.
```

Figure 4: Module predictions/forecast

```
========== Combined System Assessment ==========
Forecast for: 2026-11-03 22:00 +1000
Assessment: Routine monitoring
  Evidence: Complete
Explanation:
  - No elevated network-stress or price warning in the aligned forecasts. The renewable-share target is not met.
  - Advisory only. Critical services remain active; an authorised operator must approve operational changes. Regional forecasts do not establish site-specific safety or reserve adequacy.
Recommendations:
  - [renewable_target_shortfall] The renewable-share target is missed; review renewable availability and flexible demand. This is not evidence of total supply shortage or blackout risk.
```

*Figure 5: Final system evaluation and recommendation*

## Overall System Performance

The demand submodule's Gaussian Process model substantially outperformed persistence, achieving an average prediction error of approximately 118 MW and reducing RMSE by 61.5%. However, no other regression algorithms were evaluated, see Appendix 1.

For price-spike prediction, Random Forest was selected with a validation F2-score of 0.660 and a higher PR-AUC than Logistic Regression. On the test dataset, it achieved 35.1% precision and 64.6% recall. Although it detected all continuing spikes, it struggled with newly occurring spikes, limiting its early-warning capabilities. Additionally, spike prevalence decreased from 5.25% in training to 0.86% in testing, potentially affecting generalisation.

For renewable-generation forecasting, Gradient Boosting outperformed persistence and daily seasonal baselines, achieving an MAE of 231.65 MW, RMSE of 332.38 MW, and $R^2$ of 0.984. It reduced MAE by 59.4% compared with persistence but exhibited an average underprediction bias of 41.30 MW, potentially affecting renewable-sufficiency assessments.

Manual testing indicated that the symbolic inference layer generated recommendations as intended. However, the absence of a structured testing approach limited the assessment of its accuracy and reliability.

Overall, the hybrid system demonstrated promising performance. Demand and renewable-generation forecasting achieved substantial improvements over their baselines, while the network-stress model performed well in identifying critical conditions. However, weaker detection of high network stress and newly occurring price spikes, alongside limited end-to-end testing, highlight areas requiring further improvement.

## Project Failures

There are three main consequences of failure in the project, which should be acknowledged in the project, which are presented in Table 2.

*Table 2: Consequences of failure*

| Potential Failure | Possible Consequence | Mitigation |
|---|---|---|
| Missing or inconsistent data is treated as valid | Inappropriate recommendations that could compromise grid stability | Reject invalid data, apply a safe default operating procedure, and require manual review before generating recommendations. |
| Failure to detect a critical network event | Delayed operator response, potentially worsening network conditions | Escalate critical predictions and implement additional monitoring and alerts for potentially severe conditions. |

| | | |
|---|---|---|
| Excessive warnings or false alarms | Unnecessary interventions, operational disruption, and alert fatigue | Prioritise alerts based on severity and provide clear explanations to help operators assess recommended actions. |

These consequences must be carefully considered due to their significant societal implications, including disruptions to essential services, increased operating costs, and reduced trust in automated decision-support systems. There are also safety concerns, as inaccurate information could lead operators to make inappropriate decisions that compromise grid stability.

Compared to conventional machine-learning predictions, the symbolic layer can enforce predefined constraints and reduce the risk of errors in safety-critical systems. Its rule-based nature ensures that ML predictions cannot override explicitly defined rules and constraints. Additionally, by reasoning over a finite set of defined states, the symbolic layer can handle unfamiliar inputs according to established logical principles. However, its effectiveness depends on the completeness and accuracy of the rules within the knowledge base.

## Conclusion

The hybrid AI system successfully demonstrates the integration of machine learning and symbolic reasoning to predict NSW electricity-network conditions and generate explainable recommendations. While demand and renewable-generation forecasting performed strongly, limitations remain in detecting high network stress and newly occurring price spikes. Further testing of the combined inference layer is needed to establish its reliability. Therefore, the system should be considered a decision-support prototype rather than a replacement for human judgement.

## REFERENCES

[1]

Transgrid, 'Deep Transition Reshapes NSW's Energy Future', Transgrid. [Online]. Available: https://www.transgrid.com.au/media-publications/news-articles/deep-transition-reshapes-nsw-s-energy-future/

[2]

Australian Energy Market Operator, 'Statement of Opportunities for the National Electricity Market', Australian Energy Market Operator Limited, Technical report on electricity market reliability and investment outlook National Electricity Market Electricity Statement of Opportunities and Energy Adequacy Assessment Projection, Aug. 2026.

## APPENDICES

```
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   critical       0.83      0.90      0.86      1650

   accuracy                           0.81     10560
  macro avg       0.75      0.76      0.76     10560
weighted avg      0.82      0.81      0.81     10560

Saved model: /Users/hanskonradpujalte/Documents/Artificial-Intelligence/A2_puja0009/Misc/saved_models/network_stress_model.joblib
Saved metadata: /Users/hanskonradpujalte/Documents/Artificial-Intelligence/A2_puja0009/Misc/saved_models/network_stress_model_metada
Status: Training and evaluation complete

========== Renewable Generation | Training ==========
Model: Histogram Gradient Boosting | Horizon: 1 hour
Configuration: Previously selected in the renewable investigation notebook

Dataset partitions (target timestamps):
train   43,871 hours | 2020-09-22 01:00:00+10:00 to 2025-09-23 23:00:00+10:00
test     8,760 hours | 2025-09-24 00:00:00+10:00 to 2026-09-23 23:00:00+10:00

Test performance:
                  MAE (MW)  RMSE (MW)    R²  Bias (MW)
Gradient Boosting   231.653    332.382 0.984    -41.304
Persistence         570.208    865.823 0.893     -0.090
Daily seasonal      674.217    938.336 0.875     -1.236
Lower MAE/RMSE is better; higher R² is better. Negative bias means underprediction.
MAE improvement over persistence: 59.4%
Saved reports: /Users/hanskonradpujalte/Documents/Artificial-Intelligence/A2_puja0009/Reports/renewable
```

*Appendix 1: Model training outputs*

```
========== Demand | Training ==========
Model: Gaussian Process Regression | Horizon: 1 hour
Evaluation: Walk-forward comparison with persistence
Fitted kernel: 1.69**2 * Matern(length_scale=[2.01, 2.3, 10.3, 16.5, 2.07, 0.791, 31.2], nu=2.5) + WhiteKernel(noise_level=0.0342)

Test performance:
                MAE (MW)  RMSE (MW)  MAPE %    R²  RMSE skill
Gaussian Process   117.951    190.895   1.661 0.982       0.615
Persistence        392.633    496.084   5.337 0.882       0.000
RMSE skill is relative to persistence; higher is better.
95% prediction-interval coverage: 93.5%

Saved model: /Users/hanskonradpujalte/Documents/Artificial-Intelligence/A2_puja0009/Misc/saved_models/demand_gp.npz
Status: Training and evaluation complete
Plot: Last 14 days (close the plot window to continue if it blocks).

========== Price Spike | Training ==========
Candidates: Logistic Regression, Random Forest | Horizon: 1 hour
Selection: Highest validation F2, with PR-AUC as the tie-breaker

Dataset partitions:
train       28703 hours, 5.25% high-price, 2020-09-22 to 2023-12-31
validation   8784 hours, 2.37% high-price, 2024-01-01 to 2024-12-31
test        15144 hours, 0.86% high-price, 2025-01-01 to 2026-09-23

Validation candidates (F2 and PR-AUC use a 0-1 scale):
logistic_regression  {'C': 0.05}  F2=0.616 PR-AUC=0.413 @ 0.73
logistic_regression  {'C': 0.5}  F2=0.609 PR-AUC=0.408 @ 0.81
random_forest        {'min_samples_leaf': 5, 'max_depth': None}  F2=0.660 PR-AUC=0.577 @ 0.28
random_forest        {'min_samples_leaf': 20, 'max_depth': 12}  F2=0.653 PR-AUC=0.569 @ 0.47
Selected random_forest {'min_samples_leaf': 5, 'max_depth': None} (validation F2 0.660)

Test performance:
                    Accuracy  Precision  Recall    F2  PR-AUC  Onset recall  Continuing recall
logistic_regression     0.968      0.170   0.715 0.436   0.392         0.471              0.984
random_forest           0.987      0.351   0.646 0.553   0.475         0.324              1.000
Persistence             0.991      0.477   0.477 0.477   0.232         0.000              1.000
Always normal           0.991      0.000   0.000 0.000   0.009         0.000              0.000
Scores are on a 0-1 scale; higher is better.

Hybrid evaluation:
                 ML only  ML + rules
Any-alert recall    0.646       0.831
Onset recall        0.324       0.676
Constraint violations: 0

Saved model: /Users/hanskonradpujalte/Documents/Artificial-Intelligence/A2_puja0009/Misc/saved_models/price_spike_model.joblib
Saved reports: /Users/hanskonradpujalte/Documents/Artificial-Intelligence/A2_puja0009/Misc/price
Status: Training and evaluation complete
```

*Appendix 2: Inference layer evaluation*

## APPENDIX: AI USAGE DECLARATION

I acknowledge the use of OpenAI GPT-5.6 Sol for research and uses of code prompting in the assignment. ChatGPT was used for light paraphrasing and rewording in this report, however, all content is my own.

Full prompt outputs can be found in prompts.md under the “Misc” folder.